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INTERPLANETARY PHASE SCINTILLATION AND THE SEARCH FOR VERY LOW FREQUENCY GRAVITATIONAL RADIATION

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ABSTRACT

We report here observations of radio wave phase scintillation, using the *Viking* spacecraft having an Earth-spacecraft link very similar to that which will be used in very low frequency (VLF) gravitational wave searches. The phase power spectrum level varies by seven orders of magnitude as the Sun-Earth-spacecraft (elongation) angle changes from 1° to 175°. It is noteworthy that a broad minimum in the S band (2.3 GHz) phase fluctuation occurs in the antisolar direction; the corresponding fractional frequency stability (square root Allan variance) is $\sim 3 \times 10^{-14}$ for 1000 s integration times. A simultaneous two-frequency, two-station observation indicates that the contribution to the phase fluctuation from the ionosphere is significant but dominated by the contribution from the interplanetary medium. Nondispersive tropospheric scintillation was not detected (upper limit to fractional frequency stability $\sim 5 \times 10^{-14}$). Evidently even observations in the antisolar direction will require higher radio frequencies, phase scintillation calibration, and correlation techniques in the data processing, for detection of gravitational bursts at the anticipated strain amplitude levels $\leq 10^{-15}$.

Subject headings: gravitation — interplanetary medium

I. INTRODUCTION

Precision Doppler tracking of distant spacecraft is being investigated as a technique for the detection of bursts or trains of gravitational radiation having characteristic time scales $\sim$30–3000 s. The method proposed is to transmit from the Earth a radio signal having a very precise frequency (the uplink). This signal is coherently transponded by the spacecraft, and received again at the Earth (the downlink). What is observable at the Earth is the time series of the difference between the transmitted and received frequencies.

There have been over the years a number of suggestions that some kind of interferometry with separated free masses might become possible at radio frequencies, and thus enable observation of low frequency gravitational radiation. Braginsky and Gertsenshtein (1967) discussed the detection of periodic relative motion of bodies at solar-system-scale separations, caused by gravitational wave trains from double stars such as i Boo. R. W. Davies (1973) first derived the effect of a plane gravitational wave on the Doppler tracking record of a single spacecraft, when the Earth-spacecraft line (both uplink and downlink) lies in the plane of the wave fronts. He carefully discussed the precision of Doppler measurement, as distinguished from ranging, and how it was improving due to the hydrogen maser timekeeping then becoming operational. Anderson (1974, 1977) raised the question of interpreting spectral density distributions of frequency residuals from precision Doppler tracking of distant spacecraft. He suggested that such spectra should show characteristic features if gravitational radiation is incident on the solar system, even if that radiation were random and isotropic.

Estabrook and Wahlquist (1975) have derived the general effect on the Doppler frequency record of gravitational radiation propagating through the solar system; it can be regarded as having contributions from both (1) displacement of the Earth and spacecraft; and (2) redshift of the frequency standard used for the tracking. These indeed combine to introduce characteristic signatures in the Doppler time series (and its spectral energy distribution). For example a single burst, short compared with the round trip–light-propagation time, is seen recurring as three pulses with relative amplitudes and spacings dependent on the tracking geometry; the amplitude of fractional frequency shift $y = \Delta\nu/\nu_0$ of each is proportional to the amplitude of the propagating gravitational strain, $\Delta l/l$. Gravitational wave searches have been proposed for deep-space missions where the round trip–light-time is large, and this characteristic signature can be resolved for a reasonable range of possible gravitational wave periods (Estabrook and Wahlquist 1975; Thorne and Braginsky 1976; Estabrook and Wahlquist 1978; Douglass and Braginsky 1979).

Tentative estimates of the amplitudes, $\Delta l/l$, expected from possible astrophysical sources are $\sim 10^{-15}$ or smaller (Thorne and Braginsky 1976). Hydrogen maser timekeeping has already reached this level of stability and is expected to continue to be improved. So of more immediate concern is the careful evaluation and control of all other physical sources of phase noise. Wahlquist *et al.* (1977) and Anderson and Estabrook (1979) summarize the most important of these noise sources and, using the data of Woo *et al.* (1976), show that plasma phase scintillation associated with refractive index fluctuations in the solar wind will strongly dominate observations made within 90° of the Sun.

In the present paper, measurements of plasma phase scintillation are reported for an Earth-spacecraft link with a fairly small round trip–light-time, but otherwise very similar to that proposed for the gravitational wave search. Using the *Viking* spacecraft orbiting Mars, observations were taken during 1976.3–1978.3 covering solar elongation angles between 1° and 175°. A simultaneous two-frequency, two-station observation on an intercontinental baseline was also performed to separate "local" ionospheric phase scintillation from "global" interplanetary scintillation. In the following the results of our analysis of the single- and dual-station data are summarized, and the implication for future tracking strategies of gravitational wave searches discussed. The conclusion is drawn that such searches must be conducted while spacecraft are in a roughly antisolar direction, and that higher radio-frequency Doppler links are very desirable.

II. OBSERVATIONS AND METHOD OF ANALYSIS

The two *Viking* orbiters encountered Mars in mid-1976. Each orbiter is instrumented with an S band ($\sim$2.3 GHz) receiver and with both S and X band ($\sim$8.4 GHz) transmitters. The S and X band transmitter frequencies are coherently related to the received S band frequency, the integral ratios being 240/221 and 880/221, respectively. Thus the two frequencies transmitted by the spacecraft are themselves coherent and have the integral ratio $f_s/f_x = 240/880 = 3/11$. Schematically, an observation is as follows. A Deep Space Network (DSN) tracking station transmits an S band signal to the spacecraft (the "uplink") which is transponded—received and retransmitted—by the spacecraft as coherently related S and X band signals (the "downlinks"). The phase of the wave is distorted as it propagates through irregularities in the solar wind plasma. Cycles of received S and X band phase are counted and recorded as functions of time using the same ground timekeeping system that drives the uplink. These S and X band phase measurements form the data set for this investigation.

Our basic method of analysis isolates downlink interplanetary and ionospheric phase scintillation from other phase noise sources. The method is described by Levy (1977) and Woo *et al.* (1976); it consists of differencing the received S band phase ϕ_s and 3/11 of the received X band phase ϕ_x. Phase fluctuations that are common to the two signals and proportional to frequency cancel. These common signals are phase shifts caused by Earth-spacecraft motion, spacecraft buffeting, (nondispersive) tropospheric scintillation, clock jitter, and *any* propagation fluctuations on the common uplink. This differential phase, $\phi_\Delta(t) = \phi_s - (3/11)\phi_x$, retains scintillation contributions from the turbulent plasma on the downlink, since only plasma refractive index fluctuations are frequency-dependent, viz. $\propto$(wavelength)^2. Power spectra of ϕ_Δ then characterize $[1 - (3/11)^2]^2 \approx 0.857$ of the plasma scintillation spectrum that would have been observed at the downlink S band frequency on a one-way path from a stationary spacecraft to the Earth and with perfect clocks.

In an alternative method of analysis one simply estimates the spacecraft orbit and removes the orbital contribution to the phase observed at a single frequency. The resulting phase time series, say at S band $\phi_s(t)$, contains contributions from uplink scintillation and clock noise (multiplied by the S band transponding ratio 240/221), phase noise introduced by the transponder and spacecraft buffeting (in fact, these are apparently not seen in our data), downlink scintillation, clock noise on reception, and errors in the assumed orbit. From our viewpoint the main advantage of this analysis scheme is that it includes phase scintillation from the troposphere.

The data were all taken with the 64 m tracking antennas of the DSN (Goldstone, California; Madrid, Spain; Tidbinbilla, Australia). Data records of 1.5–12 hours duration were analyzed. Power spectra were computed by prewhitening the phase time series, Fourier transforming and squaring, postdarkening, and smoothing over adjacent spectral estimates (see Woo *et al.* 1976 for an example of a phase time series and a more complete discussion of the data processing). The spectrum is "two-sided"; that is, the variance is the integral over both positive and negative fluctuation frequencies.

In gravitational wave astronomy the emphasis is on clock stability as the ultimate criterion of achievable sensitivity. Hence it is useful to convert from the phase spectra to a measure of equivalent fractional frequency stability. A widely used figure of merit for frequency standards is the square root of the Allan variance (Barnes *et al.* 1971). The Allan variance, $\sigma_y^2(\tau)$, is the structure function of successive locally time-averaged fractional frequency fluctuations:

$$\sigma_y^2(\tau) \equiv \tfrac{1}{2}\langle[\bar{y}(t) - \bar{y}(t + \tau)]^2\rangle \,, \qquad (1)$$

where y is the instantaneous fractional frequency deviation $\Delta\nu/\nu_0$ and

$$\bar{y}(t) = \tau^{-1} \int_t^{t+\tau} y(t')dt' \,.$$

For discussing system sensitivity to pulses or bursts of gravitational radiation, the square root Allan variance of various physical sources of phase fluctuation gives a conservative measure of sensitivity—for continuous wave trans or other signals of specific frequency content it is unduly conservative (Douglass 1979).

From the definitions above, the two-sided phase power spectrum, $S_\phi(f)$, and the Allan variance are related by

$$\sigma_y^2(\tau) = \int_0^\infty S_\phi(f) f^2 \nu_0^{-2} \frac{\sin^4(\pi\tau f)}{(\pi\tau f)^2}\, df \,. \qquad (2)$$

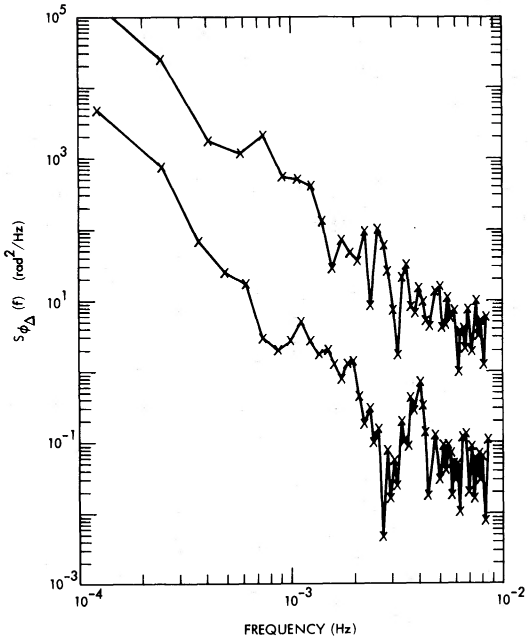

FIG. 1.—Phase power spectra in the VLF gravitational astronomy band. *Upper curve*, data at 1977 October 16 (elongation 88°); *lower curve*, data of 1978 January 20 (elongation 175°). Instrumental noise level is white with magnitude $\approx 10^{-2}$ rad^2 Hz^{-1}.

If the phase spectrum is of the form $S_\phi(f) = Af^{-a}$, equation (2) becomes

$$\sigma_y^2(\tau) = A\pi^{-2}\nu_0^{-2}\tau^{a-3}\int_0^\infty z^{-a}\sin^4(\pi z)dz \qquad (3)$$

for $2 < a < 4$.

III. RESULTS AND DISCUSSION

Figure 1 shows typical power spectra of ϕ_Δ, for elongations near 90° and 180°. In the frequency range of interest to VLF gravitational wave astronomy, the phase spectra are well characterized by a simple power law. Woo and Armstrong (1979) have analyzed these and other data in detail and show that the observations are consistent with Kolmogorov turbulence in the solar wind. Hence we adopt the spectral index $a = 8/3$ for computational purposes below.

Figure 2 is a plot of the spectral level S_{ϕ_Δ}, evaluated at 10^{-3} Hz, as a function of elongation. (In terms of the above equation, the quantity plotted is approximately $A \times 10^8$ if f is measured in Hz.) The 488 points in Figure 2 are derived from 2702 hours of dual-frequency tracking of the *Viking* orbiters (see Woo and Armstrong 1979). The right-hand axis of the figure is labeled with σ_y(1000 s), using equation (3) ($a = 8/3$). The large variations in spectral level at fixed elongation mostly reflect real time variation of the plasma turbulence rather than estimation error. As expected, the phase fluctuations minimize in the antisolar direction.

The observations in Figure 2 can be compared with model calculations for the expected dependence of interplanetary phase scintillation on elongation. From Woo (1975), the phase spectrum resulting from irregularities much larger than a Fresnel zone size is given by

$$S_\phi(f) \approx 8\pi k^2 \int_{\text{source}}^{\text{Earth}} dx \int_0^\infty dq \frac{0.033c_n^2(x)}{V(x)} \times \left\{q^2 + \left[\frac{2\pi f}{V(x)}\right]^2\right\}^{-p/2}, \qquad (4)$$

where $c_n(x)$ is the structure constant of refractive index along the line of sight, p is the spectral index of the three-dimensional spectrum ($a = p - 1$), $V(x)$ is the component of the solar wind velocity normal to the line of sight, f is the fluctuation frequency, and k is the free space wavenumber. The structure constant is a measure of the level of turbulence, and many observers, e.g., Bourgois (1969), Readhead (1971), and Armstrong and Coles (1978), have concluded that c_n varies roughly as R^{-2}, where R is the distance from the Sun.

The curve in Figure 2 was obtained by numerically integrating equation (4) for the Earth-Mars geometry assuming $c_n \propto R^{-2}$ and a constant, radially directed, solar wind velocity; the curve is normalized to agree with the data at small elongations. Aberration due to Earth and Mars orbital velocities has been ignored. The minimum in the antisolar direction is due to reduction of turbulence level along the line of sight ($c_n \propto R^{-2}$) and to the less efficient spatial wavenumber–temporal frequency transformation (the line of sight and radial wind velocity are almost parallel). This second effect causes the phase fluctuation power to be shifted toward zero frequency and out of the band of interest. The model clearly reproduces the general variation of the observed results, but the large variations in the data and the occasional systematic differences between the data and the model indicate that this simple model cannot be complete.

The spatial statistics of the phase scintillation are also of interest. On one day, 1978 March 13, we obtained 1.6 hours of simultaneous $\phi_\Delta(t)$ data at Deep Space Stations 14 (California) and 63 (Spain). The solar elongation was 121° and at the time of observation it was late afternoon in California and near local midnight in Spain. The phase autospectra S_{ϕ_Δ} observed at the two stations were similar and had spectral levels at 10^{-3} Hz of 35 rad^2 Hz^{-1} (California) and 25 rad^2 Hz^{-1} (Spain). The two $\phi_\Delta(t)$ records were digitally high-pass filtered ($f_{3\text{dB}} \approx 10^{-3}$ Hz) to remove secular trends and ensure statistical stationarity, then cross-correlated following procedures recommended by Jenkins and Watts (1978). The cross-correlation function (ccf) is shown in Figure 3. The offset of the ccf from zero time lag has the correct sense and magnitude for a pattern of phase irregularities moving outward from the Sun with solar wind speeds. (In fact, two-station phase scintillations can in principle be used to remotely sense solar wind speeds; see Woo 1977.) The correlation at zero time lag is about 0.7, so that most of the phase fluctuation power is corre-

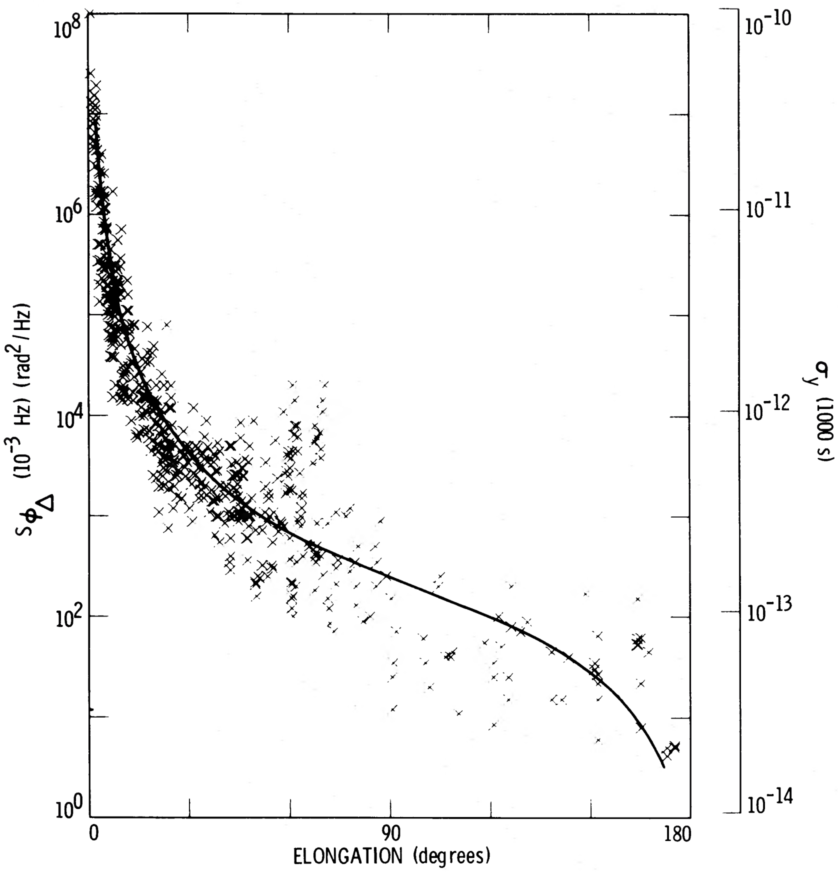

FIG. 2.—Spectral level at 10^{-3} Hz, S_{ϕ_Δ} (10^{-3} Hz), versus elongation. Right-hand scale is square root Allan variance at $\tau = 1000$ s, evaluated from eq. (3) with $a = 8/3$. Model curve through data evaluated using eq. (4).

lated over our ~10,000 km baseline. We conclude that any ionospheric contribution is secondary to that from the interplanetary medium.

For a few of the tracking passes near opposition when hydrogen maser frequency standards are employed, we have analyzed the round trip S band phase, ϕ_s, separately from the one-way differential phase, ϕ_Δ. As explained above, the main contributions to ϕ_s are from both up- and downlink scintillation (plasma and troposphere) and errors in the fitted orbit. By comparing spectra of ϕ_s and ϕ_Δ taken at the same time we can estimate the contribution to ϕ_s from the plasma and hence isolate the sum of the nondispersive contributions (principally tropospheric scintillation

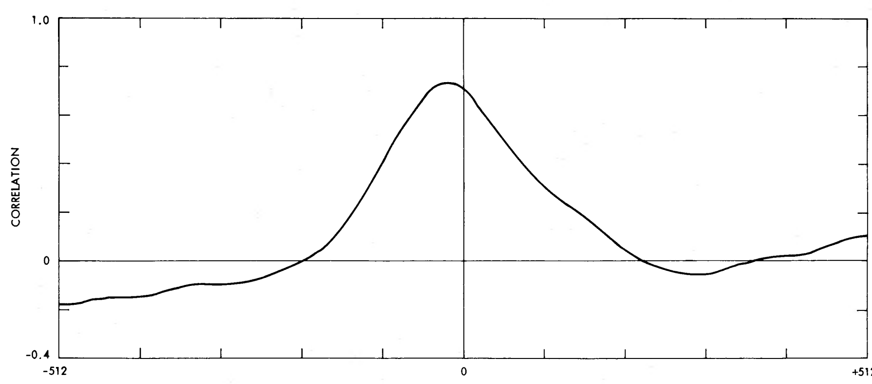

FIG. 3.—Cross-correlation function of phase data taken simultaneously at DSS 14 (California) and DSS 63 (Spain) on 1978 March 13. Solar elongation is 121°. Offset from zero time lag has correct sense and magnitude for pattern of phase irregularities convected outward from Sun with solar wind speeds.

and orbit errors). If plasma scintillation were the only source of phase fluctuations one would expect S_{ϕ_s} to be about 2.79 times larger than S_{ϕ_Δ}. This factor arises because there are scintillations on the uplink, the uplink fluctuations are multiplied by the transponder ratio 240/221, and there are downlink scintillations. If the plasma turbulence is uncorrelated on the up- and downlink then the phase variances add. Since plasma phase fluctuations are proportional to wavelength, we obtain the factor $[1 + (240/221)^4]/[1 - (3/11)^2]^2 \approx 2.79$. Alternatively, the more realistic assumption of partial correlation of the turbulence encountered on the up- and downlinks would have given a slightly larger factor.

By subtracting 2.79 times S_{ϕ_Δ} from S_{ϕ_s} for four Goldstone tracks made in early 1978 (elongations $>150°$) and attributing the difference entirely to the neutral atmosphere, we obtain values for tropospheric σ_y(1000 s) in the range 0 to 9×10^{-14}. We thus have no reliable detection of tropospheric scintillation in the present observations and adopt σ_y (1000 s) $\approx 5 \times 10^{-14}$ as a limit for nighttime desert observations in the winter. This limit can be compared with actual measurements of tropospheric scintillation on long horizontal paths over the Pacific Ocean (Thompson *et al.* 1975). The results of Thompson *et al.* need to be scaled to account for the differences in observing geometry. This scaling clearly depends on assumptions about the effective thickness of the troposphere and the distribution of turbulence with altitude, but we calculate σ_y (1000 s) $\sim 5 \times 10^{-14}$ (see also Callahan 1978).

Although the emphasis in this paper has been on the effects of interplanetary phase scintillations on the search for VLF gravitation radiation, the data shown in Figure 2 are also useful for assessing the effects on gravitation and relativity experiments (e.g., J_2) being considered for a close solar probe mission (Anderson and Lau 1978).

IV. CONCLUSIONS

We conclude that proposed VLF gravitational wave observations using coherent Doppler tracking of distant spacecraft in the antisolar direction will have S band plasma scintillation noise at the level σ_y (1000 s) $\sim 3 \times 10^{-14}$. The plasma scintillation is highly correlated on an international baseline and hence the solar wind contribution dominates the ionospheric contribution. By using an X band radio link, the plasma noise can be reduced to σ_y (1000 s) $\sim 3 \times 10^{-15}$ but tropospheric scintillation may then dominate. Evidently some combination of higher radio frequencies, phase scintillation calibration and correlation techniques in the data processing will be needed for detection of VLF bursts with strain amplitudes smaller than 10^{-15}.

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